

eralized functions - all the operations that are involved in solving for the unknown functions.

Section 2 of this paper introduces the spaces of generalized functions considered here, gives conditions when convolutions are defined for generalized functions of interest, and when products of Fourier transforms are defined. Then the convolution theorem ("exchange formula") is provided making it possible to transform convolutions into products of Fourier transforms. Section 3 gives results on existence and uniqueness of solutions of the transformed equation or system of equations. These results give conditions for identification. General results on well-posedness of the solutions are proved here in Section 3 (some were previously given in the working paper Zinde-Walsh, 2009). Section 4 provides stochastic properties of generalized functions. Section 5 gives conditions for consistent (in the topology of generalized functions) deconvolution and also for consistent non-parametric estimation of a regression function in the model (7 – 9).

2 Convolution equations in generalized functions

2.1 Spaces of generalized functions

Many different spaces of generalized functions can be defined; each may be best suited to some particular class of problems. This paper focuses on well